A Galaxy-based One Health Antimicrobial Resistance Platform

ABRomics ANTIBIORESISTANCE

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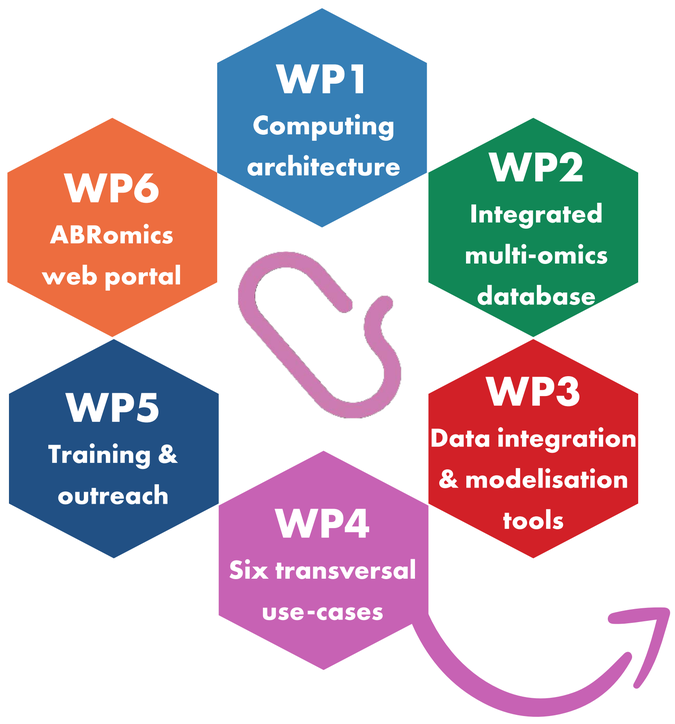
Antibiotic Microbial Resistance (AMR) is a major public health issue prioritized by international institutions. Funded by the French Priority Plan on antibioresistance (2M€; 2021-2025), we are developing a shared platform to assist AMR surveillance across human and veterinary medicine, including environmental and food isolates. It is made of :

- a repository of structured, interoperable, standardized and well-annotated multi-omics microbiological data from human, animal and environmental origins
- with an up-to-date arsenal of mathematical and bioinformatic tools to answer generic and specific research questions related to AMR.



1. Multidrug resistant bacteria
2. One Health approach
3. Meta-pan genomics for tracking AMR
4. AMR in pulmonary infection
5. AMR dissemination through aquatic env.
6. Proteomics for AMR phenotyping

An online community-driven platform to scale up and improve surveillance and research on antibiotic resistance from a One Health perspective !

ENA European Nucleotide Archive

Data brokering

Users: Clinical microbiologist, Researchers

Metadata: Sample, Isolate host, Phenotype

Sequencing Data: Reads (fastq), Consensus (fasta), Sequencing information

Metadata referential — Julie Lao

Web application interface — Romain Dallet, Raphaël Tackx, Amanda Dieuaide, Thomas Mignon, Fabien Mareuil

Database structure — Julie Lao

ABRomics web application

Data warehouse

Filesystem
- Raw data sequences
- ABR public resources
- Raw datafile results

ABRomics-DB (PostgreSQL)
- Projects and users
- Metadata
- KB on resistance genes
- WFs results

Project group and users — Upload data — Metadata validation — Results visualisation — UI to query ABRomics-DB

GALAXY CONNECTOR (django-to-galaxy) — Kenzo-Hugo Hillion, Fabien Mareuil

UseGalaxy FR server

Trimming, QC metrics, Assembly, Taxonomy, Annotation, Resistome, Virulome, MLST

Tools added & updated — Workflows added

ABRomics Platform

Galaxy workflows — Pierre Marin, Alix de Thoissy, Cléa Siguret, Bérénice Batut, Hugo Lefeuvre

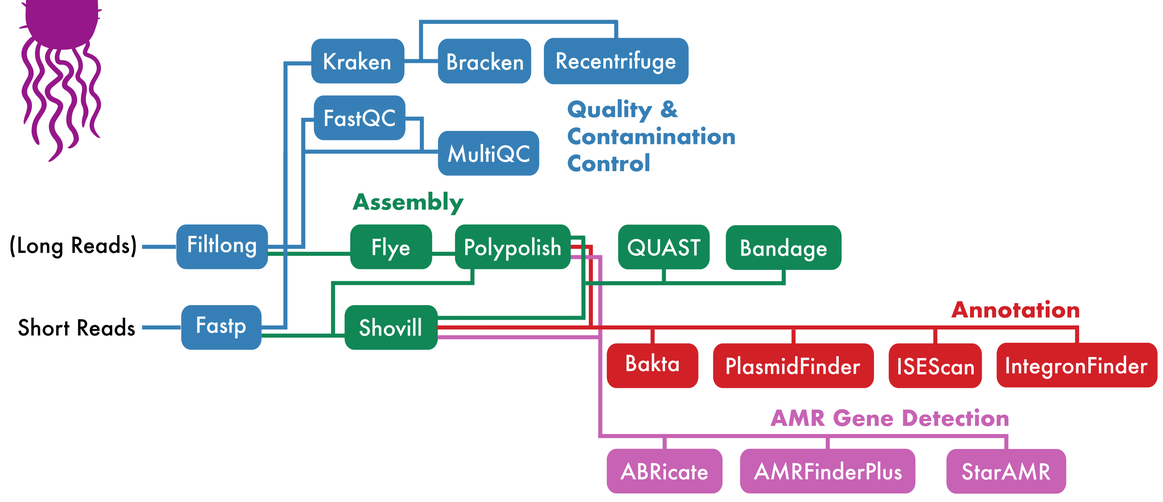
IFB core Cluster — Gildas Le Corguillé

IFB Core Cluster

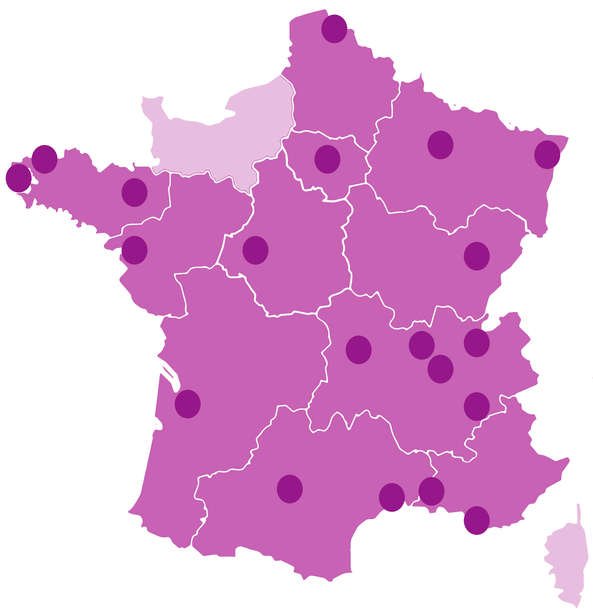
- Shared project space with right access
- Unix access to data/tools

Genomics Workflows

Galaxy FRANCE



cgMLST 2024 — Metagenomics By 2025 — Pangenomic By 2025 — OMICs By 2025



ABRomics consortium

- Effort at the national level : 43 research teams and platform!
- Multidisciplinary consortium
- Includes all National Reference Centers for antibiotic resistance
- Coordination by the French Institute for Bioinformatics (Claudine Médigue) and Institut Pasteur (Philippe Glaser)

ABRomics website

IFB INSTITUT FRANÇAIS DE BIOINFORMATIQUE — elixir FRANCE — cnrs — Inserm — INSTITUT PASTEUR — INRAE — cea — FRANCE 2030

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